University of Notre Dame

Department of Computer Science and Engineering

CSE 40567/60567: Computer Security

Assignment Date:	2/26/26
Submission Date:	
Student Name:	Mwayi Kashoko
Challenge of HW difficulty: (1 – least challenging to 5 – most challenging) 4, Vigenere cipher was the most challenging however mostly this homework was just time consuming.	

* This is the HW2 assignment which contains 100 points. You are encouraged to finish independently. Any submitted work that is copied from any source or too similar to be an independent write-up will not be given credit. Please submit this homework (as a word or PDF file including the cover page) to ND Canvas by **11:59pm on Thursday, 02/26/2026.**

Problem 1 (5 pts). Rail-Fence Cipher is a type of transposition cipher. Please decrypt the cyphertext "**CWEODITRLN**" with 3 "rails".

Note: In the Rail-Fence cipher, the plaintext is written downwards diagonally on successive "rails" of an imaginary fence, then moving up when the bottom rail is reached, down again when the top rail is reached, and so on until the whole plaintext is written out. The ciphertext is then read off in rows. For your reference, the right figure shows an example of how to use 3 "rails" for encryption.

"WE ARE DISCOVERED. RUN AT ONCE."

with 3 "rails",

```
W . . . E . . . C . . . R . . . U . . . O . . .
. E . R . D . S . O . E . E . R . N . T . N . E
. . A . . . I . . . V . . . D . . . A . . . C .
```

COLD WINTER

Problem 2 (10 pts). Please crack the Caesar cipher "**JVTWBALYZJPLUJL**" using statistical analysis. You are asked to find the key and use it for decryption (5 pts). Please also show your steps (5 pts).

COMPUTER SCIENCE

Shifted by 7

Steps posted to GitHub link

Problem 3 (25 pts). The following message is a Vigenère cipher.

```
TTEUM GQNDV EOIOL EDIRE MQTGS DAFDR CDYOX IZGZP PTAAI TUCSI XFBXY SUNFE
SQRHI SAFHR TQRVS VQNBE EEAQG IBHDV SNARI DANSL EXESX EDSNJ AWEXA ODDHX
EYPKS YEAES RYOET OXYZP PTAAI TUCRY BETHX UFINR
```

- Please find the key (5 pts) and decipher it (5 pts).
- Show your steps in detail (15 pts): i.e., using the method introduced in the class to break the cipher (references: "Lecture_CSE 40567_Spring 2026_W4_L05_Cryptography (Caesar & Vigenère Ciphers).pdf" and Assigned Reading "AR-4_CSE 40567_W4_T2-Cryptography (Caesar & Vigenère Ciphers).pdf"). More specifically, you need to show the repetitions you find (5 pts), how you establish the period and exploit index of coincidence to facilitate your estimation (5 pts), how you solve each part after you break the message into *n* parts – i.e., *n* denotes the established period (5 pts). You may write code to help solve the problem.

Note that only providing the key and decrypted text won't get full credits – i.e., you will need to show the key steps as stated above to illustrate how you solve the problem.

Letters	Start	End	Gap Length	Factors of Gap Length
DV	8	88	80	1,2,4,5,8,10, 16,20,40,80
QN	6	76	70	1,2,5,7,10,14 ,35,70
DA	25	95	70	1,2,5,7,10,14 ,35,70
PPTAAITUC	39	139	100	1,2,4,5,10,10 20,25,50,100
ED	15	105	90	1,2,3,5,6,9, 10,15,18,30, 45,90
ES	59	128	69	1,3,23,69
SN	90	107	17	1,17

Started with a period of 5 just to see and got this:

```
1: TGEEMDCIPTXSSSTVEISDEEAOEYROPTBU
2: TQODQADZTUFUQAQQEBNAXDWDYEYXTUEF
3: ENIITFYGACBNRFRNAHANESEDPAOYACTI
4: UDORGDOZASXFHHVBQDRSSNXHKEEZARHN
5: MVLESRXPIIYEIRSEGVILXJAXSSTPIYXR

               ABCDEFGHIJKLMNOPQRSTUVWXYZ
LetterFreq: 1: 11126010200010220144110110
IOC 1: 0.0625
LetterFreq: 2: 31043200000001105003301221
IOC 2: 0.0625
LetterFreq: 3: 51213211300004110212000020
IOC 3: 0.05443548387096774
LetterFreq: 4: 21032114001002201330110202
IOC 4: 0.04233870967741935
LetterFreq: 5: 10003010510210020341020420
IOC 5: 0.06451612903225806

Now the resulting alphabets are just simple Caesar Ciphers

1: Not shifted
2: Shifted by 12
3: Not shifted
4: Shifted by 25
5: Shifted by 4

Key: AMAZE
Text without formatting: THEVI GENER ECIPH ERISA METHO DOFEN CRYPT INGAL
PHABE TICTE XTBYU SINGA SERIE SOFIN TERWO VENCA ESARC IPHER SBASE DONTH
ELETT ERSOF AKEYW ORDIT EMPLO YSAFO RMOFP OLYAL PHABE TICSU BSTIT UTION

Decrypted Text: The Vigenere cipher is a method of encrypting alphabetic
text by using a series of interwoven Caesar ciphers based on the letters
of a keyword. It employs a form of polyalphabetic substitution.
```

Problem 4 (10 pts). If one-time pads are provably secure, why are they rarely used in practice?

You would have to deliver the one-time pad to the receiver which is risky. Need a one-time pad for every message sent. The one-time pad has to be the same size as the data encrypted which is not always efficient.

Problem 5 (10 pts). By using RSA algorithm, Bob wants to send Mary message "BRIGHT" (01 17 08 06 07 19) in confidence (confidentiality-assured) and authenticated (integrity-checked).

Given the following keys:

– Bob's keys: public (17, 77); private: 53
– Mary's keys: public: (37, 77); private: 13

(1) Please show how Bob will encipher the message.
e=17
n=77
d=53
m_t=[1, 17, 8, 6, 7, 19]
c=m_t^e mod n

[1, 19, 57, 41, 28, 24]

(2) Please show how Mary will decipher the received message.

e=37
n=77
d=53
m_u = [1, 19, 57, 41, 28, 24]
c=m_u^d mod n

[1, 17, 8, 6, 7, 19]

code attached in js file sent.

Note: You may leverage PowerMod Calculator to facilitate your calculation:
https://www.boxentriq.com/code-breaking/modular-exponentiation

Problem 6 (40 pts). The following message was enciphered using DES algorithm with the key of "LOVECSND" under electronic code book mode (ECB).

Ciphertext in binary format:
1100101011101101101000100110010101011111101101110011100001110011
Key: "LOVECSND" in binary format:
0100110001001111010101100100010101000011010100110100111001000100

- Please decipher it (10 pts).
- Please show key steps (20 pts). You may take references of "Lecture_CSE 40567_Spring 2026_W5_L06_Cryptography_ DES Algorithm.pdf", "Lecture_CSE 40567_Spring 2026_W5_L07_ Cryptography_ DES Algorithm (Cont).pdf" and Assigned Reading "AR-5_CSE 40567_W5_T2-Cryptography (DES Algorithm Illustration).pdf"). More specifically, you need to show the 16 generated round keys (10 pts), the output of *f* function in each iteration, *LnRn* (1<=n<=16) in each iteration (10 pts).
- Please post your code in GitHub and submit the link of your posted code (10 pts).

Note that only providing the decrypted text won't get full credits – i.e., you will need to show the key steps as stated above to illustrate how you solve the problem.

UNDIRISH

Keys in the reverse order for decryption

K1: 1010000010010010001010101111001010001000101100 01
K2: 1110000110010010001000100110110001101010010101 01
K3: 1100000010000110001000111001110101101100101000 10
K4: 1100000010100101001001000011100110010111001101 10
K5: 0100000001101100001001001011001011111100101100 0
K6: 0001000000101100010001001100110110101010011010 01
K7: 0001000000111000100011001101010001010100111100 10
K8: 0001100100001010100010000011110011011011000011 10
K9: 0001100100001000100010110101011111101110100100 011
K10: 0000101110000001100010010010101110010010111111 011
K11: 0100111101000001000010011110010000111101110111 01
K12: 0000111001000101010100010100011010101000101011 10
K13: 0000011001110001010100000111000101010001011001 11
K14: 0010010001011010010100000011111100110011100110 00
K15: 1010000000010010010100101100001100101111000111 11
K16: 1010000010010010110000101011001111100000110111 10

Function values:

f1: 011010101100111100110011011111111
f2: 001101000001001101111100101111111
f3: 110110111000010010101100001000000
f4: 101101101100000111111000101111100
f5: 011110001100010111011101010011111
f6: 110001101000101101111000000101101
f7: 100010111110110110010001110000010
f8: 001101110001101000000011010001111
f9: 101011111001001000000110101011110
f10: 011000111101000100110011000011110
f11: 100101111100011010011001110010000
f12: 101110101010000001100101111010011
f13: 010111111110011110100110010001111
f14: 001000110001011000111110110000000
f15: 101111100010000001111010000110111
f16: 010100011001101111001000110100100

All work is shown as code in the GitHub Link Here:
https://github.com/MwayiKashoko/ComputerSecurity/tree/main/Homework2